

Template for Preparing an Article

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Abstract: The availability of air conditioning systems plays an essential role in the infrastructure of higher education institutions, as it directly affects thermal comfort, the preservation of facilities, and the continuity of teaching, research, and extension activities. In this context, this study aimed to analyze the factors associated with the unavailability of air conditioning systems at the Center of Science and Technology in Energy and Sustainability (CETENS), Federal University of Recôncavo da Bahia (UFRB), and to propose improvements for maintenance management. The research was characterized as an applied case study with a qualitative approach and exploratory and descriptive objectives. Data collection was carried out through a literature review, documentary research based on the university's Institutional Development Plan, and semi-structured interviews with professionals responsible for infrastructure and maintenance management. The methodological approach was structured according to the PDCA cycle. During the planning stage, brainstorming, the Ishikawa Diagram, and the 5 Whys technique were used to identify and analyze the root causes of equipment failures. Subsequently, the GUT Matrix was applied to prioritize the identified causes, and a 5W2H action plan was developed to propose improvement actions. The results showed that the unavailability of the equipment is associated with technical, operational, and managerial factors, particularly the limited maintenance window, the centralized maintenance contract, delays in spare parts procurement, the reduced maintenance staff, and deficiencies in the campus electrical infrastructure. Based on these findings, improvement actions were proposed to strengthen the Planning and Control of Maintenance (PCM), contributing to increased equipment availability, greater process reliability, and continuous improvement of maintenance management in the institution.

Keywords: Maintenance management. Air conditioning systems. Planning and Control of Maintenance. PDCA. Quality tools.

Abbreviations: CETENS, Center of Science and Technology in Energy and Sustainability. UFRB, Federal University of Recôncavo da Bahia. PCM, Planning and Control of Maintenance. PDCA, Plan-Do-Check-Act. GTA, Technical-Administrative Management Office. NUGTEAD, Technical-Administrative Management Center. CEMAN, Engineering and Maintenance Coordination. GUT, Gravity-Urgency-Trend. SIPAC, Integrated System for Administration of Assets, Administration and Contracts.

1. Introduction

At CETENS/UFRB, air conditioning systems support teaching, research and extension activities in classrooms, laboratories and administrative sectors. According to institutional reports [5], air conditioning has been progressively expanded to improve academic activities, resulting in a significant increase in the number of air-conditioned environments. However, equipment availability remains limited due to recurring maintenance failures and

limitations in the maintenance management process.

This work aims to analyze the causes of the unavailability of CETENS/UFRB air conditioning units using quality tools, namely the Ishikawa Diagram, the 5 Whys, the GUT Matrix and the 5W2H action plan, organized according to the PDCA cycle. Based on the identified causes, improvement actions are proposed for Maintenance Planning and Control

to increase equipment availability and improve infrastructure maintenance management.

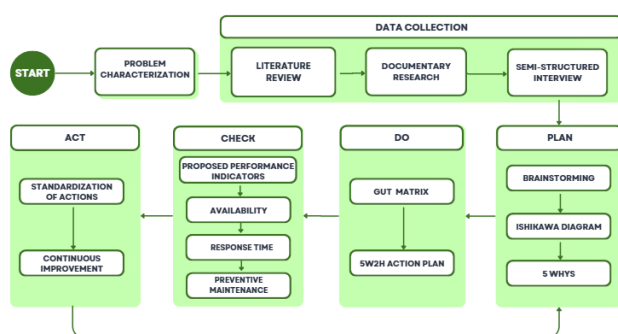
2. Methodology

The research is classified as exploratory and descriptive, adopting an applied case study with a qualitative approach focused on the maintenance management of CETENS/UFRB air conditioning systems, according to Gil [4].

Data were collected through bibliographic and documentary research and a semi-structured interview with a civil servant from the Technical-Administrative Management Center (NUGTEAD), the sector responsible for monitoring the campus infrastructure and maintenance demands. The interview data were analyzed together with the information obtained from the literature and institutional documents.

The methodological procedure was structured according to the PDCA cycle, as shown in Figure 1.

Figure 1. Methodological procedure..



In the Planning (P) stage, information obtained from the literature, institutional documents and

the interview was analyzed through brainstorming, followed by the Ishikawa Diagram and the 5 Whys technique to identify the root causes of equipment unavailability.

In Execution (D), the GUT Matrix was applied to prioritize the identified root causes, supporting the development of a 5W2H action plan. In Verification (C), performance indicators were proposed to monitor future results, while the Action (A) stage established measures for standardizing maintenance procedures, periodically reviewing the preventive maintenance plan and continuously monitoring the proposed indicators.

3. Results and discussion

The maintenance of CETENS air conditioning systems is carried out by an outsourced company contracted by UFRB under the management of the Infrastructure and Environment Coordination (CIMAN). On campus, the Technical-Administrative Management Center (NUGTEAD) monitors maintenance demands, records occurrences and communicates with CIMAN.

The process begins when teachers, technicians or users identify equipment failures. Occurrences are recorded by NUGTEAD in the institutional system and forwarded to CIMAN, which requests assistance from the outsourced company. Technical visits take place monthly,

and NUGTEAD organizes the registered requests in advance to facilitate the interventions. During these visits, inspections, filter cleaning, refrigerant gas recharging and component replacement are carried out when necessary. After the service, NUGTEAD verifies the equipment operation before closing the request.

The interviews showed that equipment often remains unavailable until the next scheduled visit due to the maintenance schedule adopted by the outsourced company. Delays may also occur because replacement parts depend on the university procurement process. In specific situations, low-cost components are purchased using the institutional corporate card or, when possible, with voluntary financial contributions.

Brainstorming also highlighted the limited maintenance capacity resulting from the reduced number of technicians available to meet the campus demand. In addition, equipment installed in classrooms and laboratories with high occupancy tends to operate under greater thermal load, accelerating component wear and increasing the occurrence of failures.

The problems identified during the brainstorming session are summarized in Table 1 and served as the basis for the subsequent root cause analysis.

Identified problem	
Equipment with insufficient cooling capacity	Reduced maintenance workforce
Prioritization of critical classrooms and laboratories	Monthly technical visits
Dependence on the outsourced maintenance company	Dependence on the university procurement process for spare parts
Long lead time for spare parts acquisition	Predominance of corrective maintenance
Preventive maintenance concentrated in specific periods	Low cooling performance
Broken condenser fan blade	Equipment fails to start
Indoor water leakage	Power outages

The investigation of the root causes was carried out using the 5 Whys technique, taking as a starting point the unavailability of air conditioning units. Through successive questions, it was possible to establish cause-and-effect relationships and identify the factors contributing to recurring failures. The results are presented in Figure 2.

Figure 2. 5 Why.



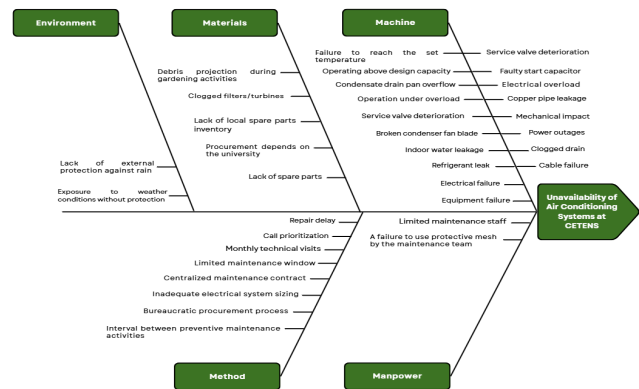
The analysis showed that equipment unavailability results from the interaction of technical, operational and managerial factors.

Technical causes included refrigerant gas leakage associated with service valve deterioration and holes in the copper tubing, electrical failures related to power outages and inadequate electrical infrastructure, and water leakage caused by clogged drainage systems. Operational factors were associated with equipment overload, the low frequency of preventive maintenance and damage caused by gardening activities near the outdoor units.

Managerial factors also contributed significantly to equipment downtime. The monthly maintenance schedule adopted by the outsourced company, the reduced maintenance team, the centralized maintenance contract and delays in the procurement of replacement parts increased the response time for corrective interventions.

Overall, the analysis showed that equipment unavailability depends not only on component failures but also on limitations in maintenance planning, resource management and infrastructure conditions. The identified root causes were subsequently organized using the Ishikawa Diagram, as presented in Figure 3.

Figure 3. Ishikawa



The Ishikawa Diagram organized the root causes identified through the 5 Whys technique into five categories (Method, Machine, Materials, Manpower and Environment), providing a comprehensive view of the factors affecting the availability of the air conditioning systems. The diagram highlights that equipment downtime results from the interaction of technical, operational, managerial and structural factors rather than isolated failures.

Among the most significant causes identified were the limited maintenance window, the centralized maintenance contract, delays in spare parts procurement, inadequate electrical infrastructure, insufficient protection of outdoor units and the predominance of corrective maintenance. Based on these findings, the GUT Matrix was applied to prioritize the identified root causes, as presented in Table 2.

Roof Case	G	U	T	GUT
R2 – Limited maintenance window	5	5	5	125
R1 – Lack of external protection against rain	4	4	4	64

R4 – Inadequate electrical system sizing	4	3	3	36
R5 – Centralized maintenance contract	3	3	3	27
R6 – Bureaucratic procurement process	3	3	2	18
R3 – Lack of risk awareness	3	2	2	12

A GUT score of 125 identified the limited maintenance window (R2) as the highest-priority issue, followed by the lack of external protection for outdoor units (R1), with 64 points. The remaining causes presented lower priority levels but were also considered relevant for improving maintenance performance.

The prioritization results supported the development of two complementary improvement plans. The first focuses on strengthening the Planning and Control of Maintenance (PCM) by improving maintenance planning and management practices. The second addresses operational and infrastructure-related issues identified during the diagnosis that cannot be solved exclusively through PCM implementation.

To support the initial implementation of the PCM, three improvement actions were proposed, as summarized in Table 3. These actions make use of existing institutional resources, including equipment records, the Integrated Asset, Administration and Contracts System (SIPAC), and the maintenance routine already adopted by the outsourced company,

minimizing implementation costs while establishing the basis for a more structured maintenance management system.

What?	Why?	Who?	How?
Standardize service order management in SIPAC	Improve intervention traceability	CIMAN	Define a standard procedure for recording service orders.
Develop the Preventive Maintenance Master Plan (PMP)	Reduce corrective failures and improve equipment reliability	CIMAN / Outsourced company	Establish preventive maintenance activities and inspection intervals.
Organize the management of critical spare parts	Reduce equipment downtime	NUGTEAD / CIMAN / Outsourced company	Implement control of critical spare parts and monitor procurement processes.
Implement periodic failure analysis and PCM review	Identify recurring causes and promote continuous improvement	NUGTEAD / CIMAN / Outsourced company	Periodically analyze maintenance records and review the preventive maintenance plan.

Note: *Where*: CETENS/UFRB. *When*: Medium term. *How much*: No additional cost.

The first action consists of standardizing the service order management process in SIPAC. Although the system is already used to record maintenance requests, the recorded information does not follow a standardized format that supports maintenance management. Standardized records, including information such as failure description, identified cause, replaced components, service time, equipment downtime and materials used, provide a reliable maintenance history that supports the

identification of recurring failures, performance monitoring and maintenance planning.

The second action consists of structuring the Preventive Maintenance Plan (PMP). Although preventive activities are already performed during specific periods, they are not yet systematically organized. Establishing a preventive plan with defined activities, inspection intervals, responsibilities and execution criteria contributes to standardizing maintenance interventions and strengthening preventive maintenance practices.

The third action proposes improving the management of critical spare parts. The analysis showed that the acquisition of replacement components is one of the main factors responsible for equipment downtime. Therefore, identifying the components with the highest replacement frequency and monitoring procurement needs can reduce replacement time and improve maintenance efficiency.

Complementary to the PCM structuring actions, additional operational and infrastructure improvement measures were proposed, as presented in Table 4.

What?	Why?	Who?	How?
Protect outdoor condenser units	Reduce weather-related deterioration	CIMAN / Administration	Install protective covers according to the manufacturer's recommendations.

Promote awareness and error-proofing practices	Prevent improper equipment use (e.g., doors left open)	Administration / CIMAN / Academic units	Conduct awareness campaigns, install warning signs and adopt automatic door-closing devices where feasible.
Inspect the electrical infrastructure	Identify failures related to electrical sizing and installation	CIMAN / Engineering	Inspect electrical circuits and compare the existing installation with the original design specifications.
Establish local emergency maintenance support	Reduce response time outside the regular maintenance schedule	CIMAN / Administration	Develop a pilot partnership with local maintenance service providers for emergency situations.
Improve the management of critical spare parts	Reduce waiting time for replacement components	CIMAN / Administration	Define a list of critical components and establish a simplified procedure for emergency procurement, in accordance with financial regulations.

Note: *Where:* CETENS/UFRB. *Time frame:* Short term for Actions 1–3; Medium term for Actions 4–5. *Estimated cost:* Budget-dependent for Actions 1, 3, 4 and 5; Low cost for Action 2.

The first action proposes installing protective covers for condenser units exposed to weather conditions. Field observations identified units subjected to direct rain and solar radiation, which may accelerate component deterioration and reduce equipment service life. Therefore, protective covers that comply with the manufacturer's recommendations while maintaining adequate ventilation are proposed.

The second action consists of adopting awareness and error-proofing measures to

encourage the proper use of air conditioning systems. The interviews indicated that keeping classroom doors open while the equipment is operating increases the cooling load and compromises system performance. Therefore, awareness campaigns, warning signs and automatic door-closing devices are proposed to reduce failures associated with improper use.

The third action proposes assessing the electrical infrastructure in areas with the highest recurrence of failures. The frequency of problems following power outages indicates the need to verify whether the electrical installations comply with the original design specifications, supporting future corrective actions.

The fourth action aims to reduce response times by establishing local emergency maintenance support. The fifth action proposes improving the management of critical spare parts by identifying frequently replaced components and monitoring procurement needs to reduce equipment downtime.

To support future decision-making and the continuous improvement of maintenance management, a set of performance indicators was proposed (Figure 4). Although SIPAC is currently used to register maintenance requests, no structured indicators were identified for monitoring maintenance performance or

evaluating the results of the proposed improvements.

Figure 4. Indicadores Propostos



Indicators of equipment availability, average service time and equipment downtime allow the operational performance of the maintenance process to be monitored. Indicators related to corrective maintenance, compliance with the Preventive Maintenance Plan (PMP) and failure recurrence support the evaluation of maintenance management performance. In addition, indicators related to the effectiveness of implemented improvements and the frequency of periodic meetings support continuous monitoring.

The Action (A) stage aims to consolidate the proposed improvements by standardizing maintenance procedures and periodically reviewing maintenance records stored in SIPAC to identify recurring failures, evaluate implemented actions and update the PMP when necessary. It is also recommended that an annual review of maintenance management be carried out before the academic recess, allowing improvements to be incorporated into the



preventive maintenance schedule. In this way, the Action (A) stage feeds back into the PDCA cycle, supporting the continuous improvement of CETENS air conditioning maintenance management.

4. Conclusion

It is concluded that the unavailability of CETENS air conditioning units was determined by the combination of technical, operational and managerial factors related to the maintenance process, whose causes were identified through the integrated application of quality tools structured in the PDCA cycle, allowing the development of proposals for the optimization of maintenance management, which may support future administrative decisions, so that the objective of the research was achieved by identifying the causes of unavailability and proposing actions aimed at improving the management of air conditioning systems maintenance.

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